



Personalized touchpoints and customer experience: A conceptual synthesis

Jakob Weidig^{*}, Marco Weippert, Christina Kuehn

Dep. VI – Marketing Department, University of Stuttgart, School of Management, Keplerstrasse 17, 70174 Stuttgart Germany

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ABSTRACT

Customers desire personalized touchpoints, and practitioners largely agree that such touchpoints constitute an essential success factor for firm and customer performance. To date, however, research has investigated the concepts of personalization and customer experience (CX) separately. Adopting a CX lens on personalization, this research draws on the findings of a conceptual synthesis of 293 articles to gain a better understanding of the effect of personalized touchpoints on CX. We find that personalization affects all CX responses, influenced by the characteristics of personalized touchpoints (touchpoint nature, customer journey stage, and personalization design). We discuss key customer contingency factors (i.e., privacy concerns and trust, proficiency, and personality traits) for the effect of personalized touchpoints on CX. This research highlights, among others, the benefits of the CX structure for personalization research to systemize its findings and the need for managers to holistically understand CX responses to personalized touchpoints.

Personalization, once limited mainly to targeted offers, now extends to the entire customer experience. This means that customers want personalization throughout their interactions with a retailer—with multiple, personalized touchpoints that enable them to allocate their time and money according to their preferences (Lindcrantz et al., 2020).

1. Introduction

Customers desire personalized touchpoints, and firms strive to eagerly address this desire (Boudet et al., 2019). More specifically, firms aim to prompt positive customer responses—that is, customer experience (CX)—by tailoring interactions (i.e., touchpoints) to individual customers' preferences and needs (i.e., personalization). At Sephora, for example, customers benefit from personalized product recommendations based on their skin tones for an enjoyable and relevant shopping experience (Pilkington & Medlock, 2020). Tesla drivers enjoy a more comfortable driving experience because their cars can proactively adapt a wide range of car settings (e.g., seat configuration, suspension) to their particular preferences (Morgan, 2021). A recent survey found that 71 % of customers expect personalized touchpoints and 76 % are disappointed when touchpoints are not tailored to them (Arora et al., 2021). A steady increase in available data and new technologies such as Internet of

Things, artificial intelligence, augmented reality/virtual reality, and biometrics further enable and propel firms to personalize touchpoints more effectively (e.g., De Keyser et al., 2021; Hoyer et al., 2020; Puntoni et al., 2021).

Despite the interplay of personalization and CX, many literature reviews and conceptual articles from the personalization (e.g., Cavdar Aksoy et al., 2021; Mehmood et al., 2023) and CX (e.g., Kranzbühler et al., 2018; Tueanrat et al., 2021) literature domains still consider the two topics separately. In addition, personalization research is fragmented within and across the marketing and information systems literature domains (Salonen & Karjalainen, 2016). Marketing research primarily focuses on isolated elements of the marketing mix, such as online advertising (e.g., Bleier & Eisenbeiss, 2015a; Boerman et al., 2017) or mobile (Tong et al., 2020) and offline (e.g., Riegger et al., 2021) channels, whereas information systems research primarily focuses on isolated technical aspects (e.g., recommender systems; Gorgoglione et al., 2019). The literature domain of CX, however, demands a holistic perspective across all marketing activities (e.g., Becker & Jaakkola, 2020), which is still absent in personalization research.

The CX literature posits that CX is subjective due to the individual customer context (e.g., Jaakkola et al., 2022) and that touchpoints need to address this context (e.g., Holmlund et al., 2020; Homburg et al., 2017; Kuehn et al., 2019). In this realm, context may comprise a diverse

^{*} Corresponding author.

E-mail addresses: jakob.weidig@bwi.uni-stuttgart.de (J. Weidig), marco.weippert@bwi.uni-stuttgart.de (M. Weippert), christina.kuehn@bwi.uni-stuttgart.de (C. Kuehn).

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set of conditional states that includes the individual, social, market, or environmental context (De Keyser et al., 2020). The CX literature domain, however, does not yet provide answers on how touchpoints should account for the individual customer context but calls for further research to address this gap (e.g., Becker & Jaakkola, 2020; Bolton et al., 2018; De Keyser et al., 2020).

The approaches, processes, and methods applied in the personalization literature can help address this gap. Therefore, personalization is a suitable tool for firms to heed the individual customer context in their activities. Only a few empirical articles touch on personalization and CX jointly. Some evaluate the effect of personalization on CX-related constructs, such as service journey quality (Jaakkola & Terho, 2021), brand experience (Tran et al., 2023), and playful CX (Lambillotte et al., 2022b), whereas others evaluate the effect of personalization on a subset of CX responses, such as emotional (Lambillotte et al., 2022a; Tyrväinen et al., 2020) and cognitive (Tyrväinen et al., 2020) responses.

Although this pioneering work confirms the relevance of personalization for CX, the overall understanding of the effect of personalized touchpoints on CX remains fragmented. We aim to bridge and combine the so-far-separate personalization and CX literature domains in a more comprehensive way. To do so, we conduct a conceptual synthesis of this research. This synthesis is based on a literature review of personalization research in the marketing and information systems literature domains through a CX lens (i.e., employing the terminology and mindset of CX). Following an analysis of 293 articles, we derive codes and structure them around key themes of the CX literature domain.

Our research provides three notable insights. First, we synthesize the current state of personalization research by adopting the CX perspective, reflecting MacInnis's (2011) conceptual goals of *revising* and *summarizing*. Doing so allows us to problematize current shortcomings of personalization research from a CX perspective (Jaakkola, 2020).

Second, we *integrate* the personalization and CX literature domains "to see previously distinct pieces as similar" (MacInnis, 2011, p. 138). We outline the conjunction of the literature domains to provide a better understanding of the effect of personalized touchpoints on CX. Specifically, we present how different characteristics of personalized touchpoints (touchpoint nature, customer journey stage, and personalization design) influence the CX responses to these touchpoints. Better understanding customer responses to personalized touchpoints adds to consumer-based strategy research (Hamilton, 2016), as we link customer responses to firms' personalization efforts. This objective is further supported by our discussion of customer-related contingency factors for the effect of personalized touchpoints.

Third, by integrating the personalization and CX literature domains we also expand the application area of personalization (Jaakkola, 2020), as our findings help both literature streams learn from each other. By assigning customer responses from personalization literature to CX responses (i.e., cognitive, emotional, behavioral, sensorial, and social) and assessing what firms can personalize (e.g., touchpoints, customer journeys, personalization design), we provide personalization research with a holistic perspective and structure to systemize its findings along the customer journey. By doing so, we overcome a major shortcoming of personalization studies that lies in its focus on single touchpoints with a rather short-term perspective, as Mehmood et al. (2023) outline. Our assessment also supplies CX scholars with a large inventory of customer responses to various personalized touchpoints and offers further evidence for the multidimensionality and multilevel nature of CX (Lemon & Verhoef, 2016). In general, a clear understanding of what the different natures of CX responses entail is currently missing. Drawing from our review of CX research, we outline how CX responses can be assigned to the different natures. As such, our research adds further empirical examples to De Keyser et al.'s (2020) building blocks from the customer perspective (CX responses). Importantly, it also contributes to CX literature, as we conceive personalization of touchpoints as playing an important role in how customers form CX.

In section 2, we introduce the focal notions of CX and personalized

touchpoints, as well as elaborate our research approach. In section 3, we first outline CX responses to personalized touchpoints and then synthesize how different characteristics of personalized touchpoints determine these CX responses. Next, we discuss which customer-related contingency factors influence CX responses to personalized touchpoints. We also derive future research directions for our findings, before concluding in section 4.

2. Points of departure

In this section, we outline our understanding of CX and personalized touchpoints and provide details on our research approach. Most definitions concur that CX reflects customers' responses to touchpoints along the customer journey. Touchpoints refer to the moment a customer encounters a firm's offerings (e.g., Homburg et al., 2017) and include, for example, the service or product and its elements, such as logo and packaging (Hoyer et al., 2020). The customer journey comprises the sum of a customer's touchpoints across a wide array of channels (Tueanrat et al., 2021) during the purchase cycle (i.e., pre-purchase, purchase, and post-purchase stages) (Lemon & Verhoef, 2016; Voorhees et al., 2017).

CX research agrees that responses have different natures, but it has not reached consensus on what these natures are. We follow the categorization of De Keyser et al. (2020), who identified cognitive, emotional, behavioral, sensorial, and social responses. Beyond the issue of differing types of natures, most research does not define what these natures entail. Therefore, to create a base for our understanding of the effects of personalized touchpoints on CX, we describe the five response natures in section 3.1.1 from established CX research. CX responses can be further characterized by their valence (i.e., positive, negative, and neutral) and magnitude (i.e., zero to low to high) (Brakus et al., 2009). As CX responses are subjective and context specific (Kranzbühler et al., 2018), firms cannot create CX but only offer touchpoints meant to elicit certain CX responses (Verhoef et al., 2009). Importantly we follow recent CX research (e.g., De Keyser et al., 2020; Kranzbühler et al., 2018), and conceive CX as non-deliberate, spontaneous, and unconscious responses to touchpoints that differ from conscious evaluative outcomes, such as customer satisfaction, loyalty, and purchase intention (Becker & Jaakkola, 2020).

Definitions of the term "personalization", by contrast, vary across study contexts (e.g., Boerman et al., 2017; Fan & Poole, 2006; Song et al., 2016). Most definitions agree that personalization is firms' proactive adaptation of touchpoints to individual customer preferences (e.g., Chellappa & Sin, 2005). It is important to note that personalization of touchpoints is initiated and managed by firms, in contrast with customization, which involves customers proactively taking part in the tailoring of touchpoints to their preferences (Arora et al., 2008; Bleier et al., 2018). To personalize touchpoints, firms first collect and analyze data on customers' preferences and behavior (Murthi & Sarkar, 2003) and then tailor respective touchpoints to the individual customer (Mehmood et al., 2023).

Technology plays a key role in personalizing touchpoints especially in digital environments (Kumar et al., 2019b), but in general, firms could personalize every touchpoint within their marketing mix (Bleier et al., 2018). The effects of personalized touchpoints on customer behavior may vary in magnitude depending on the degree to which personalization truly matches customers' preferences (Li & Unger, 2012). Importantly, although research highlights positive effects of personalization for customers and firms, it also shows negative effects, such as perceived vulnerability of customers. Scholars call this ambivalence of positive and negative customer reactions to personalization the "personalization paradox" (Aguirre et al., 2015; Aguirre et al., 2016).

Personalization literature focuses on adapting touchpoints to individual customers' preferences (Chellappa & Sin, 2005), which customers form depending on their respective contexts (Arora et al., 2008). CX research highlights the individual and subjective nature of CXs (e.g.,

Verhoef et al., 2009). The subjective nature implies that individual customers may perceive the same touchpoint in a different way. Thus, CX scholars have identified personalization as an important (design) element of the CX construct (e.g., perceived omnichannel CX (Rahman et al., 2022), service journey quality (Jaakkola & Terho, 2021), touchpoint adaptability (Holmlund et al., 2020). Despite this clear link between personalization and CX, CX literature remains vague on the role and relevance of personalization for CX, a gap our study aims to address.

By synthesizing research on personalization through a CX lens, we establish a relationship between these two concepts (MacInnis, 2011). According to Jaakkola (2020, p. 21), a *theory synthesis* is a type of conceptual work that “seeks to achieve conceptual integration across multiple theories or literature streams. Such papers offer a new or enhanced view of a concept or phenomenon by linking previously unconnected or incompatible pieces in a novel way.” Consequently, as personalization research is fragmented and reports ambiguous customer outcomes, we drew from the CX literature to structure personalization research to provide an integrated understanding. We did so with a domain-based literature review (Palmatier et al., 2018), which is a suitable tool for conceptual articles (Jaakkola, 2020). To conduct our literature review, we relied on established guidelines and processes (e.g., Snyder, 2019; see Web Appendix A for a more detailed description).

For our synthesis we used a content analysis approach (e.g., Gaur & Kumar, 2018) and created a coding template to capture the purpose of each study, its main findings, context of personalization, methodology, and variables in case of empirical research (e.g., Shamsollahi et al., 2021). We derived codes and grouped these codes into higher-order categories (Gioia et al., 2013). This process was guided by core themes from influential CX articles (e.g., Becker & Jaakkola, 2020; Lemon & Verhoef, 2016). Specifically, we followed the terminology of De Keyser et al. (2020) to structure the codes and categories. Web Appendix B provides exemplary coding results of our content analysis.

3. Toward a better understanding of the effects of personalized touchpoints on CX

From our conceptual synthesis, we derive a more comprehensive understanding of the effects of personalized touchpoints on CX and present directions for future research we consider most relevant. Fig. 1 depicts our conceptual framework. First, we discuss which CX responses

arise from personalized touchpoints. Second, we outline the key characteristics of personalized touchpoints. Finally, we explain how key customer-related contingency factors influence the effect of personalized touchpoints on CX responses.

3.1. Which CX responses arise from personalized touchpoints?

3.1.1. CX responses

As stated previously, firms adapt personalized touchpoints to customers' preferences and desires, which in turn influence CX. CX manifests itself in customer responses to touchpoints during the customer journey (Lemon & Verhoef, 2016). Therefore, to improve understanding of the effects of personalized touchpoints on CX, we investigate CX responses to personalized touchpoints. To do so, we labeled customer responses from the personalization literature as CX responses if they could be considered spontaneous and unconscious. Otherwise, we considered them evaluative outcomes. While we acknowledge that prior work in CX has not adopted this distinction (e.g., Lemon & Verhoef, 2016) between CX responses (i.e., spontaneous and unconscious) and evaluative outcomes (i.e., more deliberate and conscious) so far, nowadays it is a key issue in conceptual CX research (Becker & Jaakkola, 2020; De Keyser et al., 2020). In line with this reasoning, we distinguish, for example, between spontaneous and unconscious behavioral CX responses, such as clicks, and deliberate and conscious behavioral evaluative outcomes, such as word of mouth or purchase completion (e.g., Bleier et al., 2019; Rose et al., 2012).

Cognitive responses refer to spontaneous mental evaluations of a touchpoint associated with a customer's goal achievement and expectation (dis)confirmation (Gentile et al., 2007). Personalization literature alludes to positive cognitive responses such as perceived usefulness (e.g., Bleier et al., 2018), brand identification (Tran et al., 2020), an increase in offering relevance (Benlian, 2015; Segijn & van Ooijen, 2022), customers' decision quality (e.g., Nath & McKechnie, 2016), attitude toward the ad (Walrave et al., 2018), and positive thoughts (Grigoriou et al., 2022). At the same time, personalized touchpoints may also cause negative cognitive responses. These responses relate to privacy concerns (e.g., Bleier & Eisenbeiss, 2015b; Van Doorn & Hoekstra, 2013) when firms use personal identifiable information (Wattal et al., 2012), skepticism (e.g., Strycharz et al., 2019), perceived manipulation (Cloarec, 2020), and lower fairness perceptions (e.g., Priester et al., 2020).

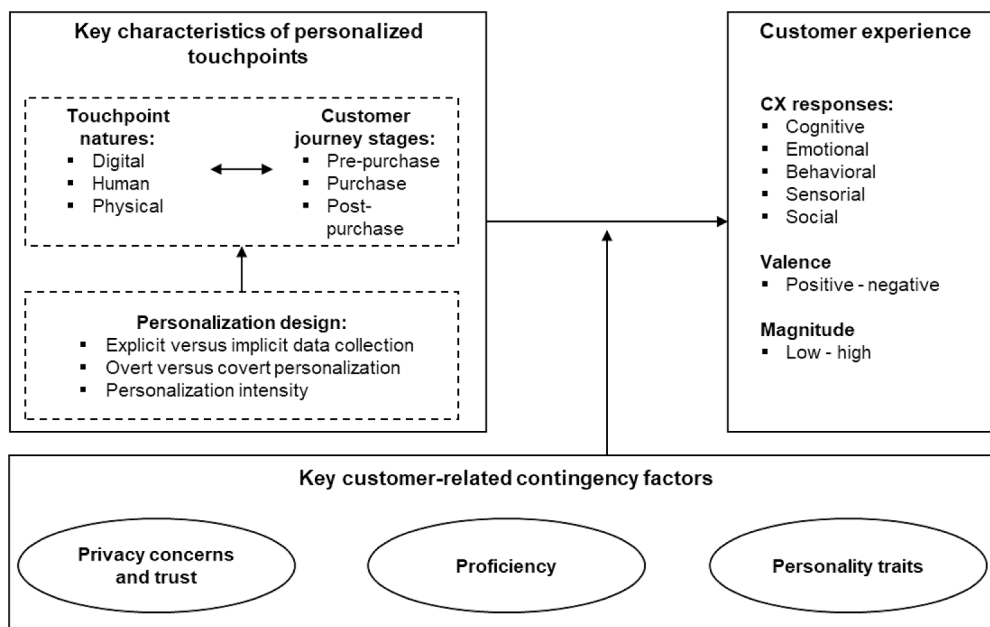


Fig. 1. Conceptual framework.

Emotional responses refer to moods and feelings evoked by a touchpoint (Gentile et al., 2007). Personalized touchpoints can lead to positive emotions, such as a feeling of being cared for (Liang et al., 2009), enjoyment (Benlian, 2015), subjective playfulness (Lambillotte et al., 2022b), and hedonic emotions (Tyrväinen et al., 2020). We also identified negative emotions, including perceived vulnerability (Aguirre et al., 2015), fear of exploitation (Riegger et al., 2021), irritation, and intrusiveness (Van Doorn & Hoekstra, 2013).

Behavioral responses refer to a customer's intended and actual activity in immediate reaction to a touchpoint (Schmitt, 1999). Personalized touchpoints cause customers to reduce information and product search behavior (e.g., Xiao & Benbasat, 2018). Especially in digital marketing communications, such as online ads and newsletters, personalized touchpoints can increase click-through intentions (e.g., Aguirre et al., 2015; Gironde & Korgaonkar, 2018), click-through behavior (e.g., Bleier & Eisenbeiss, 2015a; Kim et al., 2019), and email open rates (e.g., Wattal et al., 2012). We also identified several negative behavioral responses, such as ad avoidance (e.g., Baek & Morimoto, 2012) and reactance to personalized touchpoints (e.g., Bleier & Eisenbeiss, 2015b).

Sensorial responses refer to the perception of a touchpoint through the five senses (Schmitt, 1999). We noted few and only positive sensorial responses in the personalization literature. Personalized touchpoints can increase visual fixation on and attention to advertisements (e.g., Pfiffelmann et al., 2020; Segijn et al., 2021).

Social responses refer to how a touchpoint affects customers' relationships with others and their view of themselves (Gentile et al., 2007). The design of personalized recommendations or websites (e.g., incorporating social cues) can lead to positive responses, such as social presence (Choi et al., 2017), socialness (e.g., honesty, helpfulness; Surprenant & Solomon, 1987), and increased familiarity (Strycharz et al., 2019). By contrast, we observed customer embarrassment as a negative social response to personalized ads in situations of high social presence in public retail stores (Hess et al., 2020).

Save for sensorial responses, we identified responses of positive and negative valence for all response natures. As discussed, these CX responses vary in magnitude depending on the implementation of personalization (Li & Unger, 2012). As a result of our conceptual synthesis, we summarize the following: *Personalized touchpoints affect all five CX responses (cognitive, emotional, behavioral, sensorial, and social) with varying degrees of magnitude (i.e., zero to low to high) and valence (i.e., positive and negative).*

3.1.2. Research directions

Section 3.1.1 sheds first light not only on which CX responses emerge from personalized touchpoints but also, in more general terms, on which natures of CX responses entail which responses. Additional insights into how exactly these effects arise and how to measure them are not yet available and pose fruitful opportunities for future research. Personalization literature mainly addresses cognitive and, to a lesser degree, emotional and behavioral responses to personalized touchpoints while neglecting investigation of sensorial and social responses. CX literature reflects this research void (e.g., Becker et al., 2023; Hamilton et al., 2021). Thus, we consider investigating the effects of personalized touchpoints on sensorial and social responses a promising area for further research. In connection with this research direction, evaluating all CX responses simultaneously would be worthwhile to determine which CX responses personalized touchpoints influence most strongly (e.g., Gahler et al., 2023). Further research could also jointly consider positive and negative CX responses to personalized touchpoints. As an empirical comparison of positive and negative CX responses through, for example, a meta-analysis does not yet exist, an open question is whether the benefits of personalized touchpoints outweigh the potential negative effects. Such a meta-analysis could also help better understand the magnitude of CX responses to personalized touchpoints. In addition, future research could elaborate on the role of CX caused by personalized touchpoints for customers' evaluative outcomes, such as customer

engagement (Kumar et al., 2019a) or customer satisfaction (Becker & Jaakkola, 2020), to assess the more long-term impact of personalization.

3.2. What are key characteristics of personalized touchpoints?

In our synthesis, we disclose touchpoint nature, customer journey stage, and personalization design as key characteristics of personalized touchpoints that have an impact on CX. Touchpoint nature indicates *what* is personalized, customer journey stage *when* personalization takes place, and the personalization design *how* personalization occurs. While touchpoint nature and customer journey stage are characteristics of and thus directly linked to the touchpoint itself, personalization design takes these characteristics into account and involves several options that can be combined and then applied to each touchpoint.

3.2.1. Touchpoint natures

De Keyser et al. (2020) identify digital (e.g., app), human (e.g., service employee), and physical (e.g., stores) as constituting the predominant touchpoint natures. Importantly, while many touchpoints combine more than one nature (e.g., digital and human in a video call with a frontline service employee), we coded personalized touchpoints with respect to their predominant nature. Mehmood et al. (2023) and Scholdra et al. (2023) are, to our knowledge, among the first to investigate personalization by touchpoint nature.

Firms personalize *digital touchpoints*, such as newsletters, chatbots (e.g., Song et al., 2016), online advertising, online recommendations (e.g., Lütjens et al., 2022), and firm websites and apps (e.g., Golrezaei et al., 2014). Correspondingly, many customers expect personalization at digital touchpoints (Schweidel et al., 2022). Personalized digital touchpoints positively affect CX by improving decision quality (e.g., Tam & Ho, 2006) and increasing content relevance (e.g., Benlian, 2015). Negative effects, by contrast, include feelings of manipulation (Cloarec, 2020) and invasiveness (Gironde & Korgaonkar, 2018). The automation level at digital touchpoints allows personalization of touchpoints in real time. Current research has mostly focused on cognitive and, to a lesser degree, behavioral CX responses. While digital touchpoints enable the automatic collection and analysis of data in real time, including contextual data, customer behavior, and usage data (Kumar et al., 2019b), this data collection potentially evokes negative responses to personalization at digital touchpoints (for further details, see 3.2.3).

In contrast with digital touchpoints, *human touchpoints* such as frontline employees do not allow for automated personalization (Scholdra et al., 2023). However, they offer high flexibility, as employees can personalize human touchpoints in situ. Human touchpoints predominantly occur in a broader service context, such as retailing and hospitality. Employees personalize, for example, service content (e.g., personalized communication; Glushko & Nomorosa, 2013) and delivery (e.g., matching personalities of customer and employee; Bettencourt & Gwinner, 1996; Leischnig et al., 2018). They can directly observe CX responses to personalized touchpoints, which provides the opportunity to further personalize services instantly (Bettencourt & Gwinner, 1996). Personalization literature indicates that personalized human touchpoints enhance CX by improving the perception of services (Surprenant & Solomon, 1987) and strengthening the relationship between customers and employees (Puccinelli et al., 2009). Negative effects on CX can arise if personalized touchpoints feel disruptive to customers' goals, resulting in lower ratings for employee competence and effectiveness (Surprenant & Solomon, 1987). Not surprisingly, research has focused on emotional and social CX responses, highlighting the relationship aspect of human personalized touchpoints. Overall, research agrees that human touchpoints result in positive CX when personalization enables employees to deliver better services (Piccoli et al., 2017).

Physical touchpoints are rather static compared with digital and human touchpoints. Firms personalize the content of printed communication such as direct mail (e.g., Houston & Jefferson, 1975), offers such as physical coupons (Wierich & Zielke, 2014), and products such as

personal photographs in theme parks (Durrant et al., 2011). Many physical touchpoints (e.g., servicescape) are not easy to change and offer little opportunity to collect data. Nevertheless, new technologies, such as mobile phone tracking, facial recognition, and in-store displays, can improve data collection at physical touchpoints and allow firms to personalize them (e.g., Riegger et al., 2021) as well as synchronize on-line and offline advertising (Segijn et al., 2021). Overall, research on physical touchpoints has mostly focused on cognitive and, to a lesser degree, social CX responses. Personalizing physical touchpoints positively affects CX by creating familiarity (e.g., recognition and greeting of customers in direct mail) and improving usability (Riegger et al., 2021) and enjoyment (e.g., a fruit basket personalized to guests' preferences in a hotel room) (Glushko & Nomorosa, 2013). If personalizing physical touchpoints relies on new technologies, however, customers often perceive it as intrusive (e.g., facial recognition); these touchpoints potentially harm CX by evoking privacy concerns (Riegger et al., 2021). Public personalized touchpoints can also result in embarrassment (Strycharz et al., 2019). In summary: *The nature of personalized touchpoints determines which CX responses are triggered predominantly. Research has focused mostly on cognitive, emotional, and behavioral CX responses to personalized digital touchpoints, emotional and social CX responses to personalized human touchpoints, and cognitive and social CX responses to personalized physical touchpoints.*

3.2.2. Customer journey stages

Next, we investigated the effect of personalized touchpoints on CX across the pre-purchase, purchase, and post-purchase stages of the customer journey. In the *pre-purchase stage*, firms personalize touchpoints such as advertising (e.g., Bleier & Eisenbeiss, 2015a), recommendations (e.g., Li & Karahanna, 2015), offers, and coupons (e.g., Chellappa & Sin, 2005). Such personalized touchpoints aim to improve CX by facilitating customer decisions by increasing goal congruency between customers' shopping goals and the retailer's presentation (Murthi & Sarkar, 2003; Siebert et al., 2020). CX responses to personalized touchpoints evoked in the pre-purchase stage are mostly cognitive and behavioral in nature. Positive responses include lower decision effort (Xiao & Benbasat, 2018), higher trust (Gorgoglione et al., 2019) and click-through rates (Bleier & Eisenbeiss, 2015a), and increased attention (Maslowska et al., 2016). Negative CX responses are mostly privacy concerns (Baek & Morimoto, 2012) and reactance (Bleier & Eisenbeiss, 2015b), which are common in this stage because the relationship between the firm and the customer is often not yet strong (Aiello et al., 2020).

In the *purchase stage*, firms personalize customer services (Rust & Chung, 2006), products (Rosenbaum et al., 2021), pricing (Arora et al., 2008), and the entire transaction process (Thirumalai & Sinha, 2009). Personalized touchpoints in this stage can go as far as eliminating the classic purchase process (e.g., Amazon Go) (Hoyer et al., 2020) and aim to improve CX by making purchases more convenient and predictable for customers (Siebert et al., 2020; Thirumalai & Sinha, 2009). CX responses at personalized touchpoints in this stage are mostly cognitive in nature. Positive responses include lower perceived transaction effort (Liang et al., 2009), better decision quality (Tam & Ho, 2006), and higher trust in the firm (Coelho & Henseler, 2012). Negative CX responses include unfairness perceptions in case of personalized prices (Priester et al., 2020).

In the *post-purchase stage*, firms personalize communication addressed to customers, services delivered (Ball et al., 2006), and rewards and loyalty programs (Nastasoiu & Vandenbosch, 2019). Here, personalized touchpoints aim to improve CX by fostering perceived closeness to the firm (Song et al., 2016). Accordingly, personalized touchpoints evoke social CX responses such as higher perceived interactivity between the firm and the customer (Song & Zinkhan, 2008).

Personalization research has focused primarily on single touchpoints in the pre-purchase stage to prompt, for example, click-through behavior (e.g., Aguirre et al., 2015) or purchases (e.g., Hayes et al.,

2021). By contrast, CX literature aims to improve customer loyalty and adopts a long-term focus, as customers' perceptions of all touchpoints determine loyalty (Kuehnl et al., 2019). Furthermore, for effectively personalized touchpoints, firms may want to acknowledge the changing aims of customers along the customer journey. The focus of personalized touchpoints shifts from facilitating customer decisions before purchase, to enhancing purchase convenience, to fostering perceived closeness to the firm after purchase. Considering the entire customer journey, our synthesis indicates that firms also need to decide when to personalize touchpoints. By simplifying and improving the decision process (Murthi and Sarkar, 2003), personalizing touchpoints can enhance CX early in the pre-purchase stage. Relevant data, however, are often not available at this point, a situation known as the "cold start" problem (e.g., Kwon & Kim, 2012). Therefore, firms need to assess whether sufficient and reliable customer data are available for personalized touchpoints to contribute positively to CX. In summary: *The effect of personalized touchpoints in the different customer journey stages depends on its fit with the aims of the customers in this stage. In the pre-purchase stage, personalized touchpoints aim to facilitate customer decisions, in the purchase stage to make purchases more convenient, and in the post-purchase stage to foster perceived closeness to the firm.*

3.2.3. Personalization design

Overall, firms may personalize their touchpoints along the customer journey in different ways, but they all tailor touchpoints to the customer to a certain degree (Fan & Poole, 2006). We synthesized three overarching personalization design factors for touchpoints from the literature that may influence CX responses: the type of data collection (i.e., explicit vs. implicit) that enables personalization, the transparency of personalization (i.e., overt vs. covert), and its intensity (i.e., zero to high).

The first personalization design factor pertains to *explicit* versus *implicit data collection* (Kwon and Kim, 2012). Explicit data collection involves asking customers directly for information, such as through surveys, to gain direct insights into their preferences, interests, and intentions (Li and Karahanna, 2015). Implicit data collection entails gathering information passively from customer interactions, such as tracking and observing their website behavior or purchase history (Murthi and Sarkar, 2003). This practice offers additional context and in-depth insights into customers' implicit needs, preferences, and behaviors not expressed otherwise. Especially in the pre-purchase stage, firms often rely on implicit methods, as customers show a lesser willingness to provide information early in the customer journey (Aiello et al., 2020). By relying on both data collection practices, firms can develop rich customer data sets and derive a comprehensive understanding of their customers to personalize touchpoints more accurately and nuancedly (Aguirre et al., 2015). However, both practices come with potential negative consequences for CX. Explicit data collection imposes cognitive efforts on customers or might cause them to intentionally provide incorrect information to protect their privacy (Glushko & Nomorosa, 2013). Many customers are unaware of implicit data collection practices, which in turn leads to negative cognitive CX responses, such as privacy concerns, if a personalized touchpoint contains implicitly collected information and the customer recognizes it as such (Taylor et al., 2009). In a similar vein, implicit data collection can trigger feelings of surveillance (Riegger et al., 2021) and vulnerability, which may result in lower trust (Aguirre et al., 2015). Overall, comparing both practices, Grigorios et al. (2022) show that explicit (vs. implicit) data collection leads to significantly higher positive cognitive, emotional, and behavioral CX responses.

The choice between *overt* versus *covert personalization* relates to the transparency of the personalization practice (Mehmood et al., 2023). Overt personalization involves openly using customer data to tailor touchpoints and then explaining the reasoning for the personalization—for example, how, why, or based on what data a touchpoint adapts to an individual customer (Aguirre et al., 2015). Covert

personalization aims to create a more subtle experience, using customer data without directly revealing the personalization. Personalization research accounts for transparency by varying cues and additional information that explain how the personalization was derived (e.g., Weidig & Kuehn, 2023). For example, in the context of website personalization, Lambillotte et al. (2022a) demonstrate that for some customers, overt personalization practices reduce the negative effect of personalized touchpoints on CX response (i.e., perceived control and privacy concerns).

Research has jointly examined the combination of overt versus covert personalization and explicit versus implicit data collection but reported ambiguous results. For example, the negative effect of implicit data collection (i.e., lower click-through intentions) disappeared with overt personalization practices (Aguirre et al., 2015). By contrast, Kim et al. (2019) show that overt personalization in combination with explicit data collection leads to only beneficial CX responses (i.e., ad effectiveness). Therefore, the interplay between overt versus covert personalization and implicit versus explicit data collection is relevant to avoid negative CX responses.

Finally, *personalization intensity* describes how closely personalized touchpoints fit customers' preferences (referred to as depth (Bleier & Eisenbeiss, 2015b) and level (Boerman et al., 2017)). It reflects the types of personal data, isolated or combined, that firms use to personalize a touchpoint (e.g., only the customer's name or a combination of name, age, and interests) (De Keyser et al., 2022c). De Keyser et al., (2022b), for example, show that using customers' interests has the strongest effect on their perceptions of personalization. In general, firms can adapt the intensity of personalized touchpoints ranging from zero to low, moderate, and high. For zero intensity, firms either may not be able to personalize touchpoints because of a lack of customer data (Vesonen & Raulas, 2006) or may proactively decide against personalizing a specific touchpoint to avoid negative CX responses (Strycharz et al., 2019). The highest intensity level, by contrast, implies an exact match between a personalized touchpoint and a customer's preferences by using many different data (De Keyser et al., 2022b). Personalization literature reports ambiguous CX responses to different levels of personalization intensity. For example, higher personalization intensity can, on the one hand, cause negative CX responses, such as reactance (Bleier & Eisenbeiss, 2015b) or reduced attitudes toward the advertiser (De Keyser et al., 2022c), but, on the other hand, evoke superior positive CX responses, such as higher click-through rates (Bleier & Eisenbeiss, 2015a) or increased attitude toward the ad (Walrave et al., 2018). Too low personalization intensity can result in undermined trust in personalization and its underlying systems, such as recommendation algorithms (Yu et al., 2016). Some researchers argue that the optimal personalization intensity lies between low and moderate personalization intensity (e.g., De Keyser et al., 2022c), whereas others regard high personalization intensity as superior (e.g., Walrave et al., 2018).

Overall, we find that the valence of CX is subject not only to whether a touchpoint of a certain nature or in a certain customer journey stage is personalized but also to how firms design the personalization. Thus, we summarize the following: *The effect of personalized touchpoints on CX depends on three personalization design factors and their combination: explicit versus implicit data collection, overt versus covert personalization transparency, and no to high personalization intensity.*

3.2.4. Research directions

Our synthesis of the roles of touchpoint nature, customer journey stage, and personalization design for CX can help systemize future personalization and CX research. First, with respect to the *touchpoint nature*, existing research has predominantly analyzed the influence of single personalized touchpoints, thus lacking a comparison of the efficiency of several personalized touchpoints of different natures. Running a meta-analysis of prior research and conducting experimental studies comparing multiple personalized touchpoints might be a fruitful way to help managers identify which specific touchpoints they should

personalize to maximize positive CX and allocate resources (e.g., time, money) most efficiently. In addition, whereas prior research has focused on digital touchpoints, future studies might investigate the effects of personalized human and physical touchpoints, but also mixed-nature touchpoints (De Keyser et al., 2020) such as technology-enabled service encounters (Larivière et al., 2017). This is an evolving research area as new technologies add digital touchpoint features to human and physical touchpoints, such as smart mirrors or automated checkouts in retail stores (Hoyer et al., 2020). Finally, while new technologies are generally associated with a positive impact on CX, research also highlights negative effects, such as increasing privacy concerns (e.g., Aguirre et al., 2015). This ambiguity raises future research opportunities on how firms can best use new technologies for personalized touchpoints to increase positive and decrease negative CX responses.

Second, regarding the *customer journey stage*, research on personalization has mainly examined touchpoints in the pre-purchase stage. As discussed, however, personalized touchpoints can also enhance CX responses in the purchase and post-purchase stages. Personalized touchpoints in the post-purchase stage are especially relevant, as they can help establish and enhance long-term customer relationships. Overall, whether the efficiency of personalized touchpoints varies across customer journey stages and in which stage they might be most efficient are largely unclear. Furthermore, while personalization research agrees that customers often view data exchanges between firms and third parties as violating privacy norms (Aguirre et al., 2015; Kim et al., 2019), it has not investigated firms' need to transfer data from one personalized touchpoint to another along the customer journey (Glushko & Nomorosa, 2013). Consequently, research is necessary to understand which data flows across touchpoint natures and customer journey stages customers perceive as acceptable. Finally, exploring sequential effects of (non-)personalized touchpoints (i.e., their interplay with personalized touchpoints) and how they affect the overall CX constitutes a fruitful research area.

Third, with respect to *personalization design*, our synthesis of the three identified design factors also discloses research voids. For example, regarding explicit and implicit data collection, no study has yet compared data collection practices across touchpoints. Therefore, whether customers' perceptions of data collection practices vary across touchpoints and customer journey stages is still unclear (Aguirre et al., 2015). Likewise, future research could investigate how customers perceive varying personalization intensity levels across touchpoints and customer journey stages. For example, customers could perceive personalized human touchpoints in the pre-purchase stage more negatively than in the post-purchase stage. In this regard, assessing whether and how managers decide which touchpoints to personalize with which intensity level would also yield interesting insights for CX management research (e.g., Homburg et al., 2017). In a similar vein, future research could examine whether and for which touchpoints and under which conditions customization (i.e., customers decide for themselves how a touchpoint should be adapted to their preferences) results in more positive CX responses than personalization (e.g., Komiak & Benbasat, 2006; Sundar & Marathe, 2010).

Finally, further research is required to better understand the role of personalization transparency across different types of touchpoints and stages of the customer journey. For example, does customers' acceptance of covert personalization differ for human versus digital or physical touchpoints, and how does this evaluation change when taking the customer journey stages into account? Identifying the optimal combination of personalization design factors for each touchpoint and stage is vital to enhance positive and mitigate negative CX responses.

3.3. How do key customer-related contingency factors influence the effect of personalized touchpoints on CX responses?

CX emerges in customers' subjective contexts and are subject to individual contingencies (Verhoef et al., 2009). This means that

contextual variables related to the customer function as moderating conditions on the relationship between personalized touchpoints and resulting CX responses (Gupta et al., 2020). We identified three overarching customer-related characteristics from both personalization and CX research that influence the effect of personalized touchpoints on CX and thus act as customer-related contingency factors: (1) customers' privacy concerns and trust, (2) proficiency, and (3) personality traits (see Table 1). Identifying and aggregating contingency factors helps explain varying CX responses to personalized touchpoints more comprehensively. Existing research has focused on the moderating role of these factors with regard to the touchpoint characteristic personalization intensity. Therefore, we focus on the effect of personalization intensity on CX responses as the moderated relationship.

3.3.1. Privacy concerns and trust

Customers' privacy concerns are not just cognitive CX responses to personalized touchpoints (e.g., Bleier & Eisenbeiss, 2015b); such concerns even before experiencing a personalized touchpoint are among the most relevant contingency factors in both personalization (e.g., Baek & Morimoto, 2012; Van Doorn & Hoekstra, 2013) and CX (e.g., Bolton et al., 2014; De Keyser et al., 2020; Holmlund et al., 2020) research. Privacy concerns can increase perceived intrusiveness (i.e., emotional CX response) when personalization intensity is too high, leading to advertisement avoidance (i.e., behavioral CX response) (e.g., Baek & Morimoto, 2012). Finally, when consumer privacy concerns are high enough to make data collection impossible, firms must personalize fewer touchpoints (Schweidel et al., 2022).

Customers' trust in a firm is another relevant contingency factor in both personalization (e.g., Bleier & Eisenbeiss, 2015b; Kalaigannam et al., 2018) and CX (e.g., Bleier et al., 2019; Lemon & Verhoef, 2016) research. For example, Bleier and Eisenbeiss (2015b) find that high personalization intensity increases click-through rates (i.e., behavioral CX response) more strongly for highly trusted than less trusted firms. Thus, we summarize the following: *In situations of high trust in the firm and low privacy concerns, high personalization intensity influences CX positively. By contrast, for situations of low trust in the firm and high privacy concerns, high personalization intensity influences CX negatively.*

3.3.2. Proficiency

Studies on personalization highlight customers' proficiency with (1) data collection and personalization procedures (e.g., Komiak & Benbasat, 2006; Segijn & van Ooijen, 2022) and (2) the product category (e.g.,

Ho et al., 2011) as an additional customer-related contingency factor. CX scholars also acknowledge the relevance of customers' proficiency and past experiences in shaping their CX responses (e.g., De Keyser et al., 2020). First, prior research has shown that customers' proficiency with personalization can increase trust (i.e., cognitive CX response) in the personalized touchpoint (Komiak & Benbasat, 2006) and decrease privacy concerns (i.e., cognitive CX response) (Li & Unger, 2012). Similarly, knowledge about personalization techniques and privacy protection activities reduces negative CX responses to personalized touchpoints (Kobsa et al., 2016). In addition, scholars have shown that customers with a high degree of technological proficiency express negative cognitive CX responses (i.e., quality perceptions; Sundar & Marathe, 2010). Second, customers with a higher (lower) proficiency in a product category find a personalized product recommendation late (early) in the customer journey more useful (Ho et al., 2011). Thus, information asymmetries seem to exist between different customer groups with varying levels of proficiency (Boerman et al., 2017) that may influence the effect of personalized touchpoints on CX responses. In summary: *Customers with a high degree of proficiency with personalization experience personalized touchpoints more positively than customers with a low degree of proficiency with personalization.*

3.3.3. Personality traits

Finally, both personalization (e.g., Cavdar Aksoy et al., 2021) and CX (e.g., Holbrook & Hirschman, 1982) research highlights the role of additional personality traits in the effect of personalized touchpoints on CX responses. Capuano et al. (2015), for example, show that customer segments preferred different website designs (e.g., type of search function) depending on the big-five personality traits (i.e., neuroticism, extraversion, openness, agreeableness, conscientiousness) and thus recommend personalizing websites (i.e., digital touchpoints) on the basis of these personality traits.

Need for cognition, or the "tendency to engage in and enjoy thinking" (Cacioppo & Petty, 1982, p. 119), plays an important role in the effect of personalized touchpoints on CX responses. This moderator captures the likelihood of a customer to perform elaborate inferences of personalized touchpoints, with customers with a high need for cognition relying less on personalized recommendations than customers with a low need for cognition. Likewise, Lambillotte et al. (2022a) show that additional transparency (i.e., overt personalization) can help reduce privacy concerns (i.e., cognitive CX response) of customers with a low need for cognition. Other personality traits that negatively moderate the

Table 1
Key customer-related contingency factors.

Key contingencies	Factors	Selected references	Moderating role in the effect of personalized touchpoints
Privacy concerns and trust	■ Privacy concerns	Baek & Morimoto, 2012; Bolton et al., 2014; Li & Unger, 2012; Van Doorn & Hoekstra, 2013	■ Greater privacy concerns increase negative CX responses
	■ Trust	Bleier et al., 2019; Bleier & Eisenbeiss, 2015b; Kalaigannam et al., 2018	■ Higher levels of trust in firms increase positive and decrease negative CX responses
Proficiency	■ With personalization	Komiak & Benbasat, 2006; Li & Unger, 2012; Segijn & van Ooijen, 2022	■ Greater proficiency with personalization procedures increases positive and reduces negative CX responses
	■ With privacy protection	Boerman et al., 2017; Kobsa et al., 2016	■ Greater privacy protection proficiency reduces negative CX responses
	■ With technology	Gupta et al., 2020; Sundar & Marathe, 2010	■ Greater technology proficiency reduces negative CX responses
	■ With product category	Ho et al., 2011	■ Greater proficiency with product category reduces positive CX responses
Personality traits	■ Big-five personality traits	Capuano et al., 2015; Teeny et al., 2021	■ Personalizing digital touchpoints to big-five personality traits increases positive CX responses
	■ Need for cognition	Lambillotte et al., 2022a; Tam & Ho, 2006	■ Greater need for cognition reduces reliance on personalized recommendations
	■ Need for uniqueness	Maslowska et al., 2016	■ Higher levels of need for uniqueness increase positive CX responses
	■ Technological anxiety	Henkens et al., 2021; Kang & Namkung, 2019	■ Higher levels of technological anxiety increase negative CX responses
	■ Narcissism	Wang et al., 2023	■ Increasing personalization intensity is more efficient for consumers with high than low narcissism levels and leads to more positive CX responses

link between personalized touchpoints and CX responses are technological anxiety (Kang & Namkung, 2019) and skepticism toward advertisements (Baek & Morimoto, 2012). By contrast, narcissism (Wang et al., 2023), need for uniqueness (Maslowska et al., 2016), and innovativeness (Gironde & Korgaonkar, 2018) are customer traits that foster positive CX responses to personalized touchpoints. Overall, these personality traits influence the effect of personalized touchpoints on CX responses, and thus firms should take them into account when personalizing touchpoints.

Overall, our discussion of the contingency factors reveals that personalized touchpoints are by no means advantageous in creating positive CX for every customer. Customers differ significantly among each other and perceive personalized touchpoints subjectively (Becker & Jaakkola, 2020). With our synthesis, we can describe a personalization-prone customer segment, characterized by low privacy concerns, high trust, high proficiency, and a personality with high innovativeness and need for uniqueness. We can also outline a personalization-averse customer segment, characterized by high privacy concerns, low trust, low proficiency, and personality traits such as technological anxiety and skepticism toward ads (Fig. 2). Consequently, we expect a variety of customer segments between these two rather extreme archetypes that differ in terms of the contingency factors (e.g., Lee et al., 2011; Thomaz et al., 2020). Thus, we summarize the following: *Customers with the personality traits of narcissism, high innovativeness, and need for uniqueness react more positively to personalized touchpoints than customers with a high need for cognition, technology anxiety, and skepticism toward advertisements.*

3.3.4. Research directions

Our discussion of contingency factors can serve as a starting point for future research to better understand which customer characteristics influence the effect of personalized touchpoints on CX responses. First, privacy concerns about personalized touchpoints are likely to evolve over time as customers become more used to personalization (De Keyser

et al., 2022c). Thus, future research could conduct a longitudinal study with a sample of diverse customers to assess how evolving privacy concerns affect CX responses to personalized touchpoints over time.

Second, personalization and privacy research has identified customer segments that differ in their privacy perceptions of personalized touchpoints (Lee et al., 2011). Still unclear, however, is whether these privacy-related segments hold true when taking the different touchpoint natures and customer journey stages into account. Likewise, exploring how such segments might vary across product categories (De Keyser et al., 2022a) and products and services (Li & Unger, 2012) would be valuable. In addition, as external factors such as technological advancements become more integrated into consumers' daily lives through digital touchpoints (e.g., Alexa), firms can collect data and personalize interactions more effortlessly (Gupta et al., 2020). Further inquiry is therefore also necessary into how consumers' perception of CX resulting from technology-driven personalization evolve throughout the customer journey.

Finally, while we outline two customer segments that are either prone or adverse to personalization, exploring how these contingency factors interact with each other would be insightful to understand the personalization preferences of more customer segments. For example, does high proficiency counterbalance high privacy concerns to some degree? Similarly, a meta-analysis comparing the effect sizes of these customer-related contingency factors could prove helpful in understanding their relative importance.

4. Conclusion

4.1. Research contributions

While personalization is highly relevant for CX, concrete research insights are missing into which CX responses arise from personalized touchpoints, which characteristics of personalized touchpoints determine the CX response, and what customer-related contingency factors

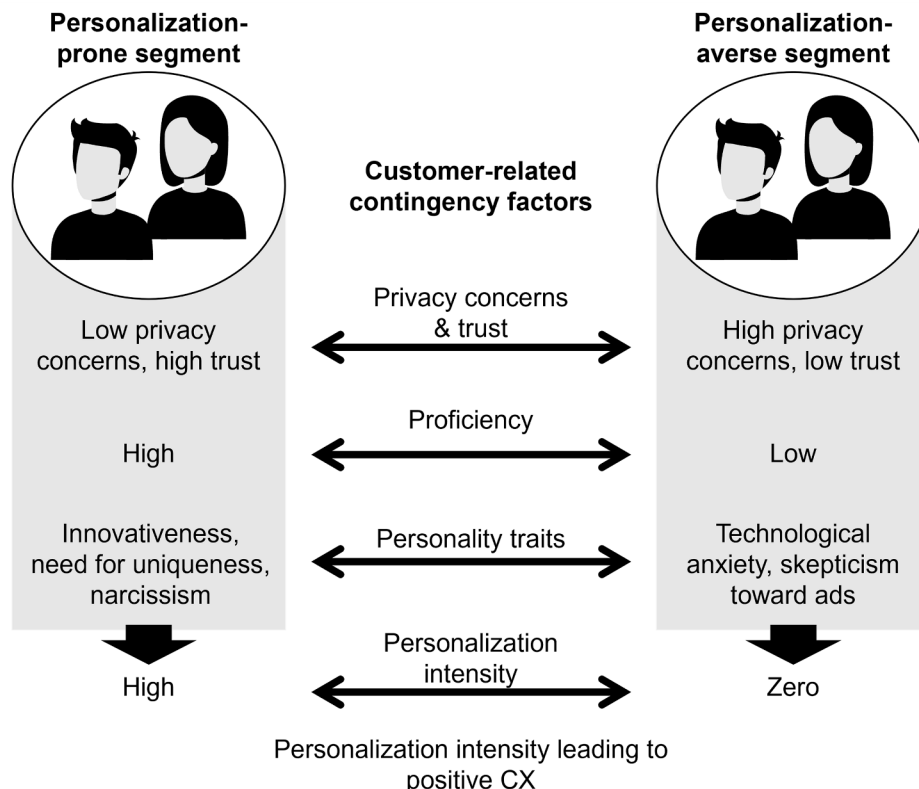


Fig. 2. Personalization intensity leading to positive CX for distinct customer segments.

Table 2
Summary of findings with research questions and managerial implications.

Summary statement	Research questions	Managerial implications
Which CX responses arise from personalized touchpoints? Personalized touchpoints affect all five CX responses (cognitive, emotional, behavioral, sensorial, and social) with varying degrees of magnitude (i.e., zero to low to high) and valence (i.e., positive and negative).	How do personalized touchpoints affect sensorial and social CX responses? Which CX responses are most strongly influenced by personalized touchpoints? When do positive CX responses outweigh the potential negative CX responses of personalized touchpoints? What is the role of CX caused by personalized touchpoints for customers' evaluative outcomes?	Managers need to be aware that personalized touchpoints do not necessarily result in a positive valence of CX responses. Negative CX responses may prevail. Managers should monitor CX responses to personalized touchpoints to adapt personalization if necessary. Managers benefit from understanding the impact of personalized touchpoints on all CX responses and adapting personalization of touchpoints accordingly.
What are the key characteristics of personalized touchpoints? The nature of personalized touchpoints determines which CX responses are triggered predominantly. Research has focused mostly on cognitive, emotional, and behavioral CX responses to personalized digital touchpoints, emotional and social CX responses to personalized human touchpoints, and cognitive and social CX responses to personalized physical touchpoints. The effect of personalized touchpoints in the different customer journey stages depends on its fit with the aims of the customers in this stage. In the pre-purchase stage, personalized touchpoints aim to facilitate customer decisions, in the purchase stage to make purchases more convenient, and in the post-purchase stage to foster perceived closeness to the firm. The effect of personalized touchpoints on CX depends on three personalization design factors and their combination: explicit versus implicit data collection, overt versus covert personalization transparency, and no to high personalization intensity.	What is the effect of personalized touchpoints of different and/or mixed natures on CX? Which personalized touchpoints are most effective in enhancing CX? How does the interplay of non-personalized and personalized touchpoints affect the overall CX? What is the effect of implicit and explicit data collection at different touchpoints? For which touchpoints is customization more effective than personalization to improve CX? How does the effect of personalization transparency vary for different personalized touchpoints?	Managers should consider the nature and customer journey stage of touchpoints when personalizing them. Managers should ensure that the design of personalized touchpoints fits the aim of customers in the customer journey stage. Managers can use three personalization design factors (data collection, personalization intensity, and personalization transparency). Managers can use three personalization design factors (data collection, personalization intensity, and personalization transparency) to optimally design their personalized touchpoints.
How do key customer-related contingency factors influence the effect of personalized touchpoints on CX responses? In situations of high trust in the firm and low privacy concerns, high personalization intensity influences CX positively. By contrast, for situations of low trust in the firm and high privacy concerns, high personalization intensity influences CX negatively. Customers with a high degree of proficiency with personalization experience personalized touchpoints more positively than customers with a low degree of proficiency with personalization. Customers with the personality traits of narcissism, high innovativeness, and need for uniqueness react more positively to personalized touchpoints than customers with a high need for cognition, technology anxiety, and skepticism toward advertisements.	How do customers' privacy concerns evolve over time and along the customer journey? How do different customer segments respond to firm-related contingency factors? How can firms understand these customer characteristics to adapt personalized touchpoints accordingly?	Managers should identify and understand customer contingency factors for individual customers to effectively personalize touchpoints. Managers should address their customers' privacy concerns and aim to build trust to improve the effect of personalized touchpoints.

influence these CX responses. Our integration of personalization and CX research resulting in a conceptual synthesis (Jaakkola, 2020) enables us to contribute to research in three main ways.

First, to provide a better understanding of personalized touchpoints this article revises and summarizes (MacInnis, 2011) the personalization literature domain through a CX lens, which is a so-far-underresearched area. We identify CX responses to personalized touchpoints and shed additional light on which characteristics of personalized touchpoints determine these responses. Specifically, we describe personalized touchpoints of different natures and along the customer journey, considering changing aims and intensity. We also outline customer-related contingency factors that influence the effectiveness of personalized touchpoints. Integrating these literature domains (MacInnis, 2011) enables us to derive a broad research agenda for personalized touchpoints (see Table 2) that may help advance both the personalization and CX literature domains.

Second, we derive implications for the CX literature domain from the personalization literature domain, and vice versa, thereby extending the application areas of both literature streams (Jaakkola, 2020). Analyzing personalization studies through a CX lens enabled us to discuss negative CX responses to personalized touchpoints. By considering both negative and positive responses, we address a major shortcoming of current CX research, which mostly highlights a positive valence of CX responses to

touchpoints.

Third, regarding the structure and the long-term orientation of personalization, we derive implications for personalization literature from CX research. Specifically, we synthesize findings on customer responses to personalization according to the structure of CX responses and evaluative outcomes. This may help better understand the “personalization paradox.” We consider the CX structure a valuable basis for future personalization research on customer responses. Gaps identified in CX research pertaining to sensorial and social CX responses, for example, also exist in personalization research. Their existence illuminates how the CX structure can serve to identify areas for future personalization research.

4.2. Managerial implications

A better understanding of the effects of personalized touchpoints can aid practitioners in several respects, as Table 2 summarizes. First, we propose that personalized touchpoints affect all CX responses to varying magnitudes and valences. Managers should therefore be aware of different valences of CX responses to personalized touchpoints, as negative CX responses (e.g., privacy concerns) may often prevail. To discern these differences, we recommend that managers closely monitor CX responses to personalized touchpoints. Ideally, they should work to

understand all five types of CX responses to personalized touchpoints holistically for each customer along the customer journey, so that they can adapt subsequent touchpoints accordingly.

Second, we propose that the effect of personalized touchpoints on CX depends on three key characteristics of personalized touchpoints: (1) the touchpoint nature, (2) the customer journey stage, and (3) the personalization design. Therefore, managers should consider all three characteristics in their design of personalized touchpoints. Furthermore, a personalized customer journey is not just a mere sequence of personalized touchpoints but requires conscious decisions on which touchpoints to personalize and how to connect them in a meaningful way to ensure a positive effect of personalized touchpoints on CX.

Third, we show that customers' privacy concerns and trust, proficiency, and personality traits are key contingency factors that influence the effect of personalized touchpoints. Therefore, managers should aim to attain data on these contingency factors for individual customers to adapt personalized touchpoints accordingly. Firms with a high share of customers who are averse to personalized touchpoints might also find it more optimal not to personalize at all. In this regard, we direct managers' attention to their control of some customer-related contingency factors to make personalized touchpoints more effective. For example, managers who uncover underlying causes of their customers' privacy concerns, such as a worry about how their data are used, can subsequently address those concerns by giving customers more input in the firm's data usage.

CRedit authorship contribution statement

Jakob Weidig: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Marco Weippert:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christina Kuehn:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

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Appendix A. Supplementary data

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Jakob Weidig is a doctoral student at the Chair of Marketing at the School of Management at the University of Stuttgart (Germany). He holds a B.Sc. and M.Sc. in Business Administration from University of Mannheim (Germany). His research focuses on Customer Experience (Management), Personalization and Customer Behavior.

Marco Weippert is a research assistant and doctoral student at the Chair of Marketing at the School of Management at the University of Stuttgart (Germany). He holds a B.A. in Business Administration from Cooperative State University Baden-Wuerttemberg (DHBW, Germany) and a M.Sc. in Management from University of Hohenheim (Germany) and HEC Management School – University of Liège (Belgium). His research focuses on the intersection of Customer Experience (Management), Customer Journey Design and Personalization.

Christina Kuehnl is a Professor of Marketing at the School of Management at the University of Stuttgart (Germany). Her research interests include Customer Experience (Management), Digital Marketing and Sales, and Product and Innovation Marketing. Her publications have appeared in, for example, the *Journal of Marketing*, *Journal of the Academy of Marketing Science*, *Journal of Retailing*, *International Journal of Research in Marketing*, *Journal of Business Research*, *British Journal of Management*, and *Psychology & Marketing*.